Graphs

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Solution of the practice quiz

1. $\mathcal{V} = \{v_0, v_1, v_2, v_3, v_4, v_5, v_6, v_7\}$ is the set of vertices.
   $\mathcal{E} = \{e_1, e_2, e_3, e_4, e_5, e_6, e_7, e_8, e_9\}$ is the set of edges.

2. The incidence function is defined as

$$\Phi : \mathcal{E} \to \mathcal{P}_2(\mathcal{V}),$$

   and maps the set of edges to the set of pair of vertices.

$$\begin{array}{ll}\Phi(e_1) = \{v_0, v_1\}\,, & \Phi(e_2) = \{v_0, v_2\}\,, \\ \Phi(e_3) = \{v_0, v_3\}\,, & \Phi(e_4) = \{v_1, v_4\}\,, \\ \Phi(e_5) = \{v_1, v_3\}\,, & \Phi(e_6) = \{v_0, v_5\}\,, \\ \Phi(e_7) = \{v_2, v_5\}\,, & \Phi(e_8) = \{v_4, v_5\}\,, \\ \Phi(e_9) = \{v_6, v_7\}\,. & \end{array}$$

3. The degree d_i of the vertex v_i is the number of incident edges. We have:

$$\begin{array}{llll}d_0 = 4, & d_1 = 3, & d_2 = 2, & d_3 = 2, \\ d_4 = 2, & d_5 = 3, & d_6 = 1, & d_7 = 1.\end{array}$$